

PROTACselect: calibrated physicochemical triage and combinatorial exploration of PROTAC design space

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ABSTRACT

Proteolysis-targeting chimeras (PROTACs) are heterobifunctional degraders assembled from a target-binding warhead, an E3-ligase ligand, and a connecting linker. The design space is combinatorial and its productive subset is small, motivating fast triage. We present **PROTACselect**, an interactive suite that assembles any warhead·linker·E3 combination, renders it in 3D, and scores ten physicochemical descriptors against problem-domain objective functions (weight vectors plus hard feasibility filters). Score thresholds are *calibrated* to the physicochemical distribution of 10,728 published degraders from PROTAC-DB, and each metric is reported with its percentile in that reference set. An Explorer mode sweeps hundreds of combinations (parametric PEG and alkyl linkers) and ranks them, with a Pareto view of likeness versus molecular weight. We emphasise, prominently, what the tool does *not* do: it models the ligand, not the ternary complex, and its index reflects consistency with known-degrader chemistry — *not* validated degradation potency or therapeutic efficacy. PROTACselect is therefore best understood as an honest developability-triage and design-exploration front-end for the chemistry half of degrader design.

Keywords: PROTAC · targeted protein degradation · induced proximity · cheminformatics · beyond-Rule-of-5 · design-space exploration

1 Introduction

PROTACs hijack the ubiquitin–proteasome system: rather than inhibiting a protein, a bifunctional molecule recruits an E3 ubiquitin ligase to a protein of interest, forming a ternary complex that tags the target for degradation [1,2]. Because activity is event-driven and catalytic, degraders can reach targets long considered undruggable [3]. But a PROTAC is defined by three interchangeable parts, and with even a modest parts list the number of assemblable molecules is large while the fraction that forms a productive ternary complex is small. Practitioners therefore need rapid triage that separates the trivially undevelopable from the plausible, before committing to synthesis or to expensive structure-based modelling.

PROTACselect addresses the tractable, honest part of this problem — the physicochemistry of the assembled chimera — while being explicit about the boundary it does not cross.

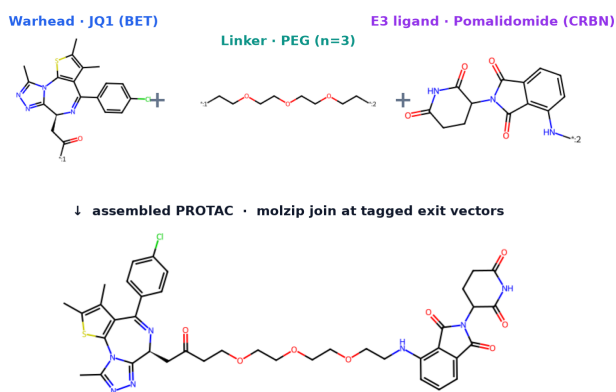


Figure 1. Component-to-assembly. A warhead, linker and E3 ligand — each carrying tagged exit-vector atoms (*:1, *:2) — are fused with RDKit *molzip* into one degrader. Example: JQ1 · PEG(*n*=3) · pomalidomide (CRBN).

2 Methods

Assembly. Warheads, E3 ligands and linkers are stored as SMILES with tagged exit vectors (mapped dummy atoms). Assembly joins matching labels via RDKit *molzip* (Fig. 1). Linkers are either fixed motifs or generated parametrically — PEG(*n*=1–12) and alkyl(C2–12).

Tier-1 descriptors. Ten RDKit descriptors are computed on the assembled molecule: molecular weight, cLogP, topological polar surface area (TPSA), H-bond donors/acceptors, rotatable bonds, aromatic rings, fraction *sp*³, ring count and synthetic accessibility. Each is mapped through a trapezoidal desirability function to a colour band and combined into a weighted 0–100 index.

Domains as objectives. A problem domain is an objective function: a weight vector over the ten descriptors plus optional hard filters. A CNS domain caps TPSA and H-bond donors; a BCL-xL / apoptosis domain *requires* a VHL handle, encoding the platelet-sparing rationale of clinical degraders such as DT2216 [8]. Selecting a domain reweights the index and gates feasibility.

Calibration. PROTACs occupy beyond-Rule-of-5 space [7], where hand-set drug-likeness cut-offs mislead. We therefore calibrate every threshold to the empirical descriptor distribution of 10,728 assembled degraders from PROTAC-DB [4] (Fig. 2): lower-is-safer metrics take green/red edges at the 75th/95th percentiles; the index thus measures consistency with known-degrader space. Each reported value carries its percentile in this reference set.

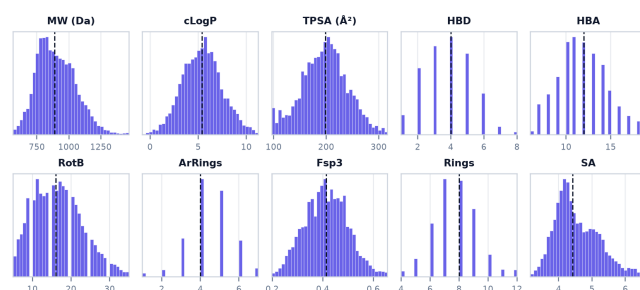


Figure 2. Physicochemical space of 10,728 PROTAC-DB degraders (dashed line: median). These distributions define PROTACselect's calibrated bands and per-metric percentile context.

3 Results

Hand-set thresholds were systematically too strict. Calibration revealed that intuition-based cut-offs would flag much of real PROTAC space as poor (Fig. 3). The starkest case is TPSA: a hand-set red edge of 180 Å² sits below the median (≈199 Å²) of published degraders. Rotatable bonds and molecular weight show the same bias. Recalibration moves the edges to where degraders actually live.

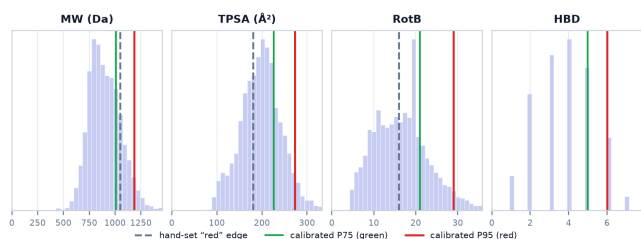


Figure 3. Hand-set “red” edges (dashed) versus data-calibrated 75th/95th-percentile edges (green/red) on four descriptors. For TPSA and rotatable bonds the hand-set edge lies left of the bulk — it would reject the majority of real PROTACs.

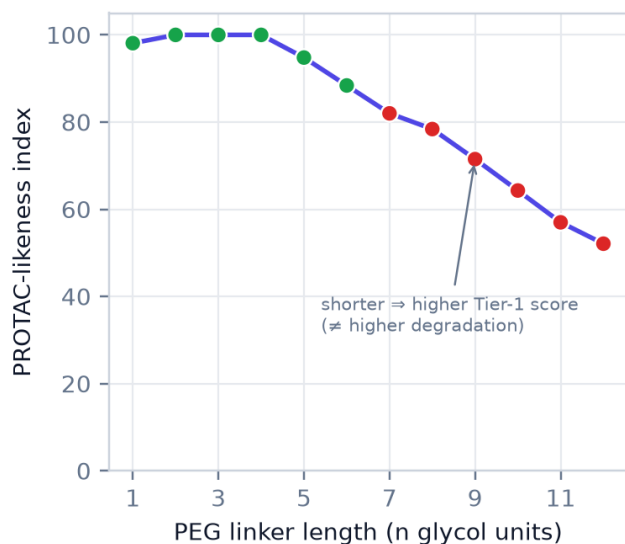


Figure 4. Tier-1 index versus PEG linker length (JQ1-pomalidomide). Developability declines monotonically with length — a caution: the productive-geometry optimum for *degradation* is a Tier-2 property this score does not capture.

Exploration and trade-offs. The Explorer sweeps every linker×E3 combination for a chosen warhead and domain (310 combinations, <1 s) and ranks them. Plotting likeness against molecular weight exposes the trade-off directly and its Pareto frontier (Fig. 5). Domain hard filters behave as intended: for a BCL-xL warhead under the apoptosis domain, exactly the VHL combinations pass and all others are gated out.

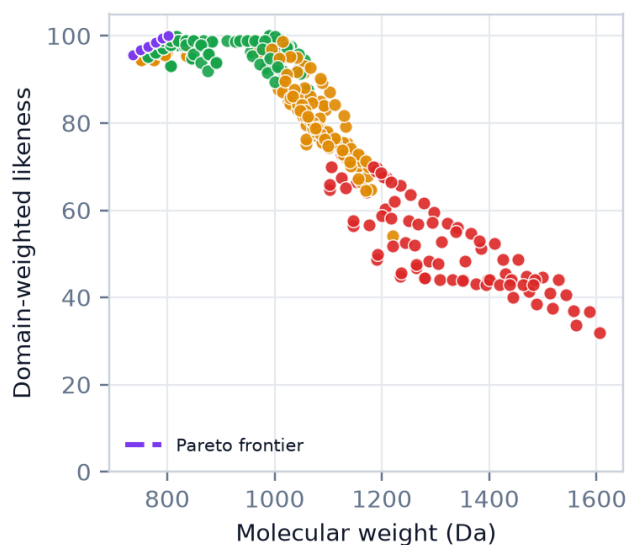


Figure 5. Explorer sweep for enzalutamide under a hormone-driven oral domain: 310 candidates by domain-weighted likeness versus molecular weight, coloured by verdict (grey = domain-blocked). Dashed: Pareto frontier.

4 Limitations

These are first-order, not caveats in fine print. (i) **The ligand is modelled, not the biology.** There is no ternary complex, cooperativity, or lysine/ubiquitination geometry — the very features that determine whether a PROTAC degrades. (ii) **The index is typicality, not efficacy.** It measures how consistent a molecule is with known-degrader chemistry; a high-scoring molecule may be biologically inert. (iii) **Nothing is validated against a measured outcome** (DC_{50} , D_{max} , permeability, oral bioavailability). (iv) Static descriptors such as TPSA miss the chameleonic behaviour that governs beyond-Rule-of-5 permeability [9]. (v) Advisory tips optimise developability and can, if followed naively, harm degradation (e.g. shortening a linker). (vi) Tier-2 — degradation potency and ternary geometry — is out of scope here.

5 Availability

PROTACselect is a Python/RDKit engine with a Flask + 3Dmol.js interface. The reference dataset derives from PROTAC-DB [4], whose terms restrict redistribution; calibration files are regenerated locally. Roadmap: stratified calibration against oral/clinical degraders; a Tier-2 degradation model reported with uncertainty; and protein/ternary-complex context.

References

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